

hour near the start of spring and are adjusted backward in autumn.)

New earth time. New earth time (NET) is another way for measuring the time of day. NET was introduced on September 15, 1999 by Mark Laugesen from Auckland. The rights to the name New Earth Time (or NET) and phrase '360 degrees of time' are novel and owned by degree NET Ltd.

In NET, the day is split into 360 NET degrees, each NET degree is split into 60 NET minutes, and each NET minute is split into 60 NET seconds. One NET degree is therefore equivalent to four standard minutes, and one standard hour is equivalent to 15 NET degrees.

NET is equivalent to the UTC read from a 24-hour analogue clock as the clockwise angle past midnight of the hour hand. For example, noon is 180°0'0" NET, and at that time the hour hand is pointing straight down forming a 180° angle when seen from the top, at midnight. A full circle is 360 degrees and one NET day.

Analogue clock

The analogue clock code definition is the same as described in the 'Mouse Tracking Calendar and Clock in HTML Using JavaScript' software article published in EFY August 2008 issue. The analogue clock tracks the mouse cursor and moves wherever the cursor goes.

One important thing in this project is locking and unlocking movement of the analogue clock. This is achieved by 'Lock' checkbox that is controlled via JavaScript function *Static()*, which checks the behaviour of the checkbox as well as motion of the mouse cursor on screen. If it is not checked, it will return function *Dynamic()*, or else it will

return 'false' to the program as per the code given below:

```
!document.getElementById("Static").checked?Dynamic():"return false";
```

The function *Arrange()* takes the time as new parametric input *new Date(<parameters>)* as given below:

```
function Arrange()
{
time=new Date(new Date().getTime())
+ (document.getElementById("tzo").options[document.getElementById("tzo").selectedIndex].value*3600*1000));
secs= time.getTimeSeconds();
sec=-1.57+Math.PI*secs/30;
mins=time.getTimeMinutes();
min=-1.57+Math.PI*mins/30;
hr=time.getHours();
```

Using new *Date()* native keyword, the selected country's time zone is sent as a dynamic parameter to *getTime()* keyword, which calculates the total time. The time is then extracted using a standard time extraction method like *new Date().getHours()*. Each hand then moves precisely according to the current time of the selected country. The movement is governed by the seconds, minutes and hours functions.

Digital clock design. The *DsBrd()* wholly takes care of all other functions seen on the dashboard, other than Analogue Clock. At the outset, the date is extracted as discussed above, a variable *f* is defined which catches the local time, as per the offset values, which is then sent to generate the Base clock.

The Swatch time is generated from the code given below:

```
b=((((tt/86400000)%1)+(1/24)).toString().substring(0,7));
```

where, the *tt: getTime* parameter is divided by 86,400,000 then modulo by 1, and then added to reciprocal of 24. It is

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

nothing but total number of seconds in a day (86,400), converted into milliseconds, 24-hour time is reciprocated and added, and up to seven decimal digits are extracted from it, and displayed on the screen.

Another variable *ee* is defined in the code for extracting time parameters for NET:

```
ee=(tt/86400000*360);
with(Math)
{
ed=floor((ee*1)%360);
em=floor((ee*60)%60);
es=floor((ee*360)%1);
}
```

As NET needs to be displayed in degrees, minutes and seconds, the resultants are modulo by 360 (for 360 degrees), and the result shown is in degrees.

The minute will be multiplied for 60 times and modulo by 60 to get in minute format.

The seconds are multiplied by 360 and then modulo by 1, for each second.

Each of these extracted outputs is then displayed on the screen.

The program was tested on Windows 7 running Internet Explorer 9.0. It was also found working well on Chrome version 23.0.1271.97 but may not work on Mozilla Firefox 24.0. ●

The author is a B.Tech in electronics and communications engineering who likes coding in JavaScript. He has a keen interest in space technology (satellite communication) and mobile communication



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